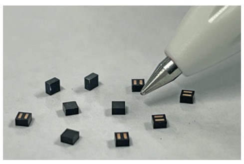
and artificial intelligence that demand higher storage and performance. It starts up about 70% faster because it uses a quicker setup speed called HS-G1 Rate A instead of the slower traditional speed. It also makes data more secure. It can better handle and protect important security data like user passwords and quickly cleans up any deleted security data safely.

Kioxia Corporation
https://www.kioxia.com

World's smallest light source device for 3D sensors

SCIVAX Corporation has developed Amtelus, a light source device for 3D sensors. It is the smallest light source device, reducing its footprint—ten times smaller and half the thickness of prior models. Amtelus reduces the space needed for light-emitting components, enabling use in devices

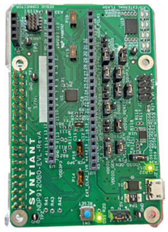


like AR glasses and smartphones. The reduction in size allows for the use of multiple modules to improve sensing precision. Amtelus integrates lenses with the light source in a resin package using nanoimprint technology, making it more affordable than traditional products. The device is suitable for other wearable technologies, including smartwatches and fitness trackers

SCIVAX Corporation
https://www.scivax.com/en

AI battery chip for more power and longer life

Eatron Technologies, in collaboration with Syntiant, has launched the AI-powered battery management system on chip. The AI-BMS-on-chip increases battery capacity by 10% and extends battery life by 25%. It comes equipped with pre-trained AI models that provide assessments of battery health, charge status, and remaining life. The system achieves battery power through estimations of charge levels and battery health, includes early detection of

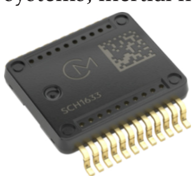


potential issues via predictive diagnostics for safe operation, and promotes longer battery life by managing health and usage. A notable feature is its real-time analysis and decision-making capabilities on the device itself. Designed for integration, the AI-BMS-on-chip is suitable for a range of battery-powered devices, including light mobility solutions and industrial and consumer electronics.

Eatron Technologies and Syntiant
https://eatron.com and
https://www.syntiant.com

Highly accurate measurement sensors for ADAS

Murata has launched the SCH1633-D01, a micro-electromechanical system (MEMS) sensor. The sensor is designed for automotive applications such as autonomous driving, driver-assistance systems, inertial navigation, vehicle

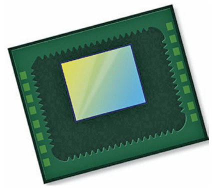


stability control, and camera or headlight alignment. The MEMS sensor is suitable for zonal architecture, supporting subsystems including GNSS integration, chassis control, and vehicle attitude sensing. It is primarily used in the central vehicle inertia measuring unit, ensuring signal integrity across all vehicle subsystems even in challenging environments. This integration simplifies system architecture and facilitates cost reductions.

Murata
https://www.murata.com

Smallest footprint sensor for facial recognition

OMNIVISION has introduced the OV0TA1B monochrome (mono)/infrared CMOS image sensor. This sensor is the first to accommodate the 3mm module Y dimension, making

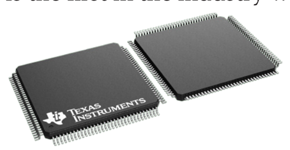


it suitable for smaller notebook computers, webcams, and IoT devices. The sensor is a low-power device designed for artificial intelligence (AI)-enabled human presence detection, facial authentication, and always-on applications. IoT device developers benefit from its low power consumption and sensitivity for continuous monitoring applications like traffic and people counting. Security system integrators can use this sensor in access control and security systems that require facial recognition and presence detection.

OMNIVISION
https://www.ovt.com

Real-time MCUs for fast fault detection

Texas Instruments (TI) has introduced a series of real-time microcontrollers for automotive and industrial applications. The TMS320F28P55x series is the first in the industry with an



integrated neural processing unit (NPU). This series improves performance by offloading neural network model execution from the main CPU to the NPU, reducing latency by five to ten times compared to software implementations. The NPU can also learn and adapt to various environments, ensuring fault detection accuracy of over 99% and enabling more informed decisions at the edge. IoT and edge computing developers creating smart devices for energy management, environmental monitoring, and smart grids benefit from the microcontroller's on-device neural processing, enabling rapid, adaptive decision-making in real time.

Texas Instruments (TI)
https://www.ti.com